Linear equations in two variables — Part 2 answer key

#	ID	Answer	Difficulty
2.1	55d9f495-d3f8-4300-aadb-4beb4b5365d2	13.75, 55/4	Medium
2.2	803e7a6a-e220-4562-862c-82f2d02fcb7f	D	Medium
2.3	4e5c1cee-ad79-4a5a-8828-24762395ae21	A	Medium
2.4	2764f11e-ac8f-4a9c-b6b0-db45ba9e4e48	C	Medium
2.5	b0ba00cd-ad0e-4693-864f-2a0283cf64f7	C	Medium
2.6	7840df29-5b0a-4ae4-9fb1-c8f458684f0b	B	Medium
2.7	9ac48e4f-1444-4974-8049-572f31e5ed81	A	Medium
2.8	d85fc288-4098-4fba-a899-571c900d8231	C	Medium
2.9	021851-DC	D	Medium
2.10	ac7b345b-6b45-4f69-945c-5e7e16ca8909	B	Medium
2.11	3567f63b-262e-4925-aa9a-31ab8da38bd7	D	Medium
2.12	efb2f017-fd60-4d55-aa5b-bf7aa344524d	A	Medium
2.13	02a67ccb-75a5-4787-89d5-0516399af231	0.25, 1/4	Medium
2.14	ab826e08-7891-4957-94db-0152ef740dff	C	Medium
2.15	b4b428b9-5d13-4880-8289-c67386f8fc40	C	Medium
2.16	e7a189d8-ea3e-4bf1-b0c2-bb03d3f29337	D	Medium
2.17	df2106f5-60f0-4169-a44d-b3223cb59788	C	Medium
2.18	eb660cfe-6d9e-465b-a9a3-9b5a5ae60195	A	Medium
2.19	6b671d8f-d733-464f-b0d9-9d825d5fd66c	49	Medium
2.20	eb3d5dbd-4820-4685-bc1e-0710105586f6	4.51, 451/100	Medium

2.21	d739e9dd-c317-4c86-831a-07375a627a3d	9	Medium
2.22	10508e20-26b1-4e5b-8b38-3ab2f8695a5f	A	Medium
2.23	e4e2c39c-307d-411e-a829-b1a79fa0bee3	46	Medium
2.24	1ede31ff-90cd-4739-bd83-5ba88c2b02cb	B	Medium
2.25	0d9d21e3-8ce9-45b7-b90e-60ee11a2b37c	C	Medium
2.26	f751d621-537a-4175-801e-a9a856b72c00	.3846, 5/13	Medium
2.27	f33b0055-46c2-4d92-8dbc-03f057ed1886	D	Medium
2.28	08073-DC	C	Medium
2.29	c9e742be-9c93-4d44-8ef4-0c15f7fff17f	C	Medium
2.30	2f6fe99e-a285-4b30-b61f-c4864a95e6e6	14.66, 14.67, 44/3	Medium
2.31	02dc7dfd-b457-45d1-8655-2bcfcc4dfc17	D	Medium
2.32	f7d44282-a880-4219-88e0-2d034cd8f06b	1.636, 18/11	Medium
2.33	029477-DC	C	Hard
2.34	43b52d01-006d-4dbc-b7a8-3187b73ccbce	A	Hard
2.35	cdc3a36c-aceb-4d6e-99f2-44a62cd1b9b1	40	Hard
2.36	029556-DC	A	Hard
2.37	8cda2479-6500-45ad-b2fe-a4863fff3443	C	Hard
2.38	658ecb3f-b141-4d7a-abcb-105abef728e2	A	Hard
2.39	ee187260-bfb5-410e-bb49-e34aec865b7b	1.2, 6/5	Hard
2.40	eb64b8f6-731a-4137-9ed2-358ab19f106c	A	Hard
2.41	7e535b93-6192-4963-87af-547e39bbdfcf	D	Hard
2.42	2af9fdfa-146c-4add-84b4-cb5bbf516dc2	C	Hard
2.43	06cf762f-e23b-41cd-9a4f-cba279a432ff	189/5, 37.8	Hard
2.44	c6aceeba-b507-44a2-bc31-9beb10096c93	D	Hard
2.45	10934-DC	A	Hard
2.46	0f40e0e9-fc67-443e-989b-f56c60365045	210	Hard

2.47	80b79c80-6158-42f6-b5e5-0c0a19031fce	D	Hard
2.48	029626-DC	?	Hard
2.49	d4043654-05ea-4067-80ff-0c1687134c44	5	Hard
2.50	2b8c0dff-e373-422a-a150-3f281e461330	0.25, 1/4	Hard
2.51	0741234a-eaa5-4593-8dae-f0666e86a222	B	Hard
2.52	83c4d031-e8f1-4415-8c6c-33d11d65e85b	33	Hard
2.53	b7ba7dde-8eb4-42fb-a134-796966b15f0d	D	Hard
2.54	070614-DC	?	Hard
2.55	045840-DC	D	Hard
2.56	8f4b667c-7537-4495-ba4d-4549d1e62f85	B	Hard
2.57	155657c1-cb62-489d-b3be-d87bbbf5aada	24	Hard
2.58	029469-DC	B	Hard
2.59	070612-DC	?	Hard
2.60	070613-DC	?	Hard
2.61	02e0a325-790e-4a20-956c-c646b0e6a227	59/9, 6.555, 6.556	Hard
2.62	a44626e2-3239-48f1-90f6-145e3b8b2c54	C	Hard
2.63	3e65e30b-6f1f-4e34-9ac7-2f2019988ed3	D	Hard
2.64	6e2c4ed8-a7d5-4552-bdd7-42223af298ea	.5, 1/2	Hard
2.65	029630-DC	?	Hard

Rationales

▼ 2.1 — 13.75, 55/4

The correct answer is $\frac{55}{4}$. In the *xy*-plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a y-intercept of $0, b$. Applying the distributive property to the right-hand side of the given equation yields $y = \frac{27}{4}x + \frac{15}{4} + 7x$. Combining like terms yields $y = \frac{55}{4}x + \frac{15}{4}$. This equation is in the form $y = mx + b$, where

$m = \frac{55}{4}$ and $b = \frac{15}{4}$. It follows that the slope of the graph of $y = \frac{1}{4}27x + 15 + 7x$ in the xy-plane is $\frac{55}{4}$. Note that 55/4 and 13.75 are examples of ways to enter a correct answer.

▼ 2.2 — D

Choice D is correct. The equation that defines line t in the xy-plane can be written in slope-intercept form $y = mx + b$, where m is the slope of line t and $0, b$ is its y-intercept. It's given that line t has a slope of $-\frac{1}{3}$. Therefore, $m = -\frac{1}{3}$. Substituting $-\frac{1}{3}$ for m in the equation $y = mx + b$ yields $y = -\frac{1}{3}x + b$, or $y = -\frac{x}{3} + b$. It's also given that line t passes through the point $9, 10$. Substituting 9 for x and 10 for y in the equation $y = -\frac{x}{3} + b$ yields $10 = -\frac{9}{3} + b$, or $10 = -3 + b$. Adding 3 to both sides of this equation yields $13 = b$. Substituting 13 for b in the equation $y = -\frac{x}{3} + b$ yields $y = -\frac{x}{3} + 13$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This equation defines a line that has a slope of 9, not $-\frac{1}{3}$, and passes through the point $0, 10$, not $9, 10$.

Choice C is incorrect. This equation defines a line that passes through the point $0, 10$, not $9, 10$.

▼ 2.3 — A

Choice A is correct. It's given that a neighborhood consists of a 2-hectare park and a 35-hectare residential area and that the total number of trees in the neighborhood is $3{,}934$. It's also given that the equation $2x + 35y = 3{,}934$ represents this situation. Since the total number of trees for a given area can be determined by taking the number of hectares times the average number of trees per hectare, this must mean that the terms $2x$ and $35y$ correspond to the number of trees in the park and in the residential area, respectively. Since $2x$ corresponds to the number of trees in the park, and 2 is the size of the park, in hectares, x must represent the average number of trees per hectare in the park.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 2.4 — C

Choice C is correct. It's given that a total of 2 squares each have side length r. Therefore, each of the squares has perimeter $4r$. Since there are a total of 2 squares, the sum of the perimeters of these squares is $4r + 4r$, which is equivalent to $24r$, or $8r$. It's also given that a total of 6 equilateral triangles each have side length t. Therefore, each of the equilateral triangles has perimeter $3t$. Since there are a total of 6 equilateral triangles, the sum of the perimeters of these triangles is $3t + 3t + 3t + 3t + 3t + 3t$, which is equivalent to $63t$, or $18t$. Since the sum of the perimeters of the squares is $8r$ and the sum of the perimeters of the triangles is $18t$, the sum of the perimeters of all these squares and triangles is $8r + 18t$. It's given that the sum of the perimeters of all these squares and triangles is 210. Therefore, the equation $8r + 18t = 210$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.5 — C

Choice C is correct. A line in the xy-plane can be represented by an equation of the form $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the line. It's given that line s passes through the point $0, 0$. Therefore, the y-coordinate of the y-intercept of line s is 0. It's also given that line s is parallel to the line represented by the equation $y = 18x + 2$. Since parallel lines have the same slope, it follows that the slope of line s is 18. Therefore, line s can be represented by the equation $y = mx + b$, where $m = 18$ and $b = 0$. Substituting 18 for m and 0 for b in $y = mx + b$ yields the equation $y = 18x + 0$, or $y = 18x$. If line s passes through the point $4, d$, then when $x = 4$, $y = d$ for the equation $y = 18x$. Substituting 4 for x and d for y in this equation yields $d = 184$, or $d = 72$.

Choice A is incorrect. This is the y-coordinate of the y-intercept of the line represented by the equation $y = 18x + 2$.

Choice B is incorrect. This is the slope of the line represented by the equation $y = 18x + 2$.

Choice D is incorrect. The line represented by the equation $y = 18x + 2$, not line s, passes through the point $4, 74$.

▼ 2.6 — B

Choice B is correct. It's given that line k is defined by $y = 7x + \frac{1}{8}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line defined by this equation in the xy-plane and b represents the y-coordinate of the y-intercept of this line. Therefore, the slope of line k is 7. It's also given that line j is perpendicular to line k in the xy-plane. Therefore, the slope of line j is the opposite reciprocal of the slope of line k. The opposite reciprocal of 7 is $-\frac{1}{7}$. Therefore, the slope of line j is $-\frac{1}{7}$.

Choice A is incorrect. This is the opposite reciprocal of the y-coordinate of the y-intercept, not the slope, of line k.

Choice C is incorrect. This is the y-coordinate of the y-intercept of line k, not the slope of line j.

Choice D is incorrect. This is the slope of a line that is parallel, not perpendicular, to line k.

▼ 2.7 — A

Choice A is correct. When the graph of an equation in the form $Ax + By = C$, where A, B, and C are constants, is shifted down k units in the xy-plane, the resulting graph can be represented by the equation $Ax + By + k = C$. It's given that the graph of $27x + 33y = 297$ is shifted down 5 units in the xy-plane. Therefore, the resulting graph can be represented by the equation $27x + 33y + 5 = 297$, or $27x + 33y + 165 = 297$. Subtracting 165 from both sides of this equation yields $27x + 33y = 132$. The y-intercept of the graph of an equation in the xy-plane is the point where the line intersects the y-axis, represented by the point $0, y$. Substituting 0 for x in the equation $27x + 33y = 132$ yields $270 + 33y = 132$, or $33y = 132$. Dividing both sides of this equation by 33 yields $y = 4$. Therefore, if the graph of $27x + 33y = 297$ is shifted down 5 units, the y-intercept of the resulting graph is $0, 4$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the y-intercept of the graph of $27x + 33y = 297$ shifted up, not down, 5 units.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.8 — C

Choice C is correct. It's given that when line n is graphed in the xy-plane, it has an x-intercept of $-4, 0$ and a y-intercept of $0, \frac{86}{3}$. The slope, m, of a line can be found using any two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the

points -4, 0 and $0, \frac{86}{3}$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{\frac{86}{3} - 0}{0 - -4}$, or $m = \frac{43}{6}$. Therefore, the slope of line n is $\frac{43}{6}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the slope of a line that has an x-intercept of $\frac{86}{3}, 0$ and a y-intercept of $0, -4$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.9 — D

Choice D is correct. Since a 160-pound adult uses 200 calories for every 30 minutes of hiking, then the same adult uses $200h$ calories after hiking for h 30-minute periods. Similarly, the same adult uses $150b$ calories after bicycling for b 30-minute periods. Therefore, the equation $200h + 150b = 1{,}900$ represents the situation where a 160-pound adult uses a total of 1,900 calories from hiking for h 30-minute periods and bicycling for b 30-minute periods. It's given that the adult completes 1 hour, or 2 30-minute periods, of bicycling. Substituting 2 for b in the equation $200h + 150b = 1{,}900$ yields $200h + 150(2) = 1{,}900$, or $200h + 300 = 1{,}900$. Subtracting 300 from both sides of this equation yields $200h = 1{,}600$. Dividing both sides by 200 yields $h = 8$. Since h represents the number of 30-minute periods spent hiking and there are 2 30-minute periods in every hour, it follows that the adult will need to hike for $\frac{8}{2}$, or 4 hours to use a total of 1,900 calories from bicycling and hiking.

Choice A is incorrect and may result from solving the equation $200h = 1{,}900$. This represents 0 30-minute periods bicycling instead of 2. Choice B is incorrect and may result from solving the equation $200h + 150 = 1{,}900$. This represents 1 30-minute period of bicycling instead of 2. Choice C is incorrect. This may result from determining that the number of 30-minute periods the adult should hike is 8, but then subtracting 2 from 8, rather than dividing 8 by 2, to find the number of hours the adult should hike.

▼ 2.10 — B

Choice B is correct. It's given that line k is defined by $y = \frac{17x}{7} + 4$. For an equation of a line written in the form $y = mx + b$, m is the slope of the line and b is the y-coordinate of the y-

intercept of the line. It follows that the slope of line k is $\frac{17}{7}$. It's also given that line j is parallel to line k in the xy-plane. Since parallel lines have equal slopes, line j also has a slope of $\frac{17}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the y-coordinate of the y-intercept of line k, not the slope of line j.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.11 — D

Choice D is correct. It's given that the equation $80S + 90C = 1{,}120$ represents this situation, where S is the number of square tokens won, C is the number of circle tokens won, and $1{,}120$ is the total number of points the tokens are worth. It follows that $80S$ represents the total number of points the square tokens are worth. Therefore, each square token is worth 80 points. It also follows that $90C$ represents the total number of points the circle tokens are worth. Therefore, each circle token is worth 90 points. Since a circle token is worth 90 points and a square token is worth 80 points, then a circle token is worth $90 - 80$, or 10, more points than a square token.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of points a circle token is worth.

Choice C is incorrect. This is the number of points a square token is worth.

▼ 2.12 — A

Choice A is correct. An equation representing the linear relationship between x and y can be written in slope-intercept form $y = mx + b$, where m is the slope of the graph of the equation in the xy-plane and $(0, b)$ is the y-intercept. The slope, m, can be calculated using two ordered pairs, x_1, y_1 and x_2, y_2, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the ordered pairs $(-6, 65)$ and $(6, 29)$ from the table for x_1, y_1 and x_2, y_2, respectively, in this formula yields $m = \frac{29 - 65}{6 - (-6)}$, which is equivalent to $m = \frac{-36}{12}$, or $m = -3$. Substituting -3 for m in the formula $y = mx + b$ yields $y = -3x + b$. Substituting the point $(-6, 65)$ into this equation yields $65 = -3(-6) + b$, or $65 = 18 + b$. Subtracting 18 from both sides of this equation yields $47 = b$. Substituting 47 for b in the equation $y = -3x + b$ yields $y = -3x + 47$. Adding $3x$ to both sides

of this equation yields $3x + y = 47$. Multiplying both sides of this equation by 3 yields $9x + 3y = 141$.

Choice B is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $9(-6) + 3(65) = 3$, or $141 = 3$, which is false.

Choice C is incorrect. Substituting the point $-6, 65$ from the table into this equation yields $3\text{-}6 + 965 = 141$, or $567 = 141$, which is false.

Choice D is incorrect. Substituting the point $-6, 65$ from the table into this equation yields $3\text{-}6 + 965 = 3$, or $567 = 3$, which is false.

▼ 2.13 — 0.25, 1/4

The correct answer is $.25$. It's given that the topsoil and crushed stone Isabel ordered for her garden cost a total of $\$ 641$. It's also given that the equation $24.5x + 24.75y = 641$ represents the relationship between the number of cubic yards of topsoil, x, and the number of tons of crushed stone, y, that Isabel ordered. Since x represents the number of cubic yards of topsoil ordered, $24.5x$ represents the total cost, in dollars, of the topsoil, and the cost per cubic yard of topsoil is $\$ 24.50$. Similarly, since y represents the number of tons of crushed stone ordered, $24.75y$ represents the total cost, in dollars, of crushed stone ordered, and the cost per ton of crushed stone is $\$ 24.75$. Therefore, a ton of crushed stone cost Isabel $24.75 - 24.50$, or 0.25, more dollars than a cubic yard of topsoil. Note that .25 and 1/4 are examples of ways to enter a correct answer.

▼ 2.14 — C

Choice C is correct. It's given that line r is perpendicular to line p in the xy-plane. This means that the slope of line r is the negative reciprocal of the slope of line p. If the equation for line p is rewritten in slope-intercept form $y = mx + b$, where m and b are constants, then m is the slope of the line and $0, b$ is its y-intercept. Subtracting $18x$ from both sides of the equation $2y + 18x = 9$ yields $2y = -18x + 9$. Dividing both sides of this equation by 2 yields $y = -9x + \frac{9}{2}$. It follows that the slope of line p is -9. The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of -9 is $\frac{-1}{-9}$, or $\frac{1}{9}$. Thus, the slope of line r is $\frac{1}{9}$.

Choice A is incorrect. This is the slope of line p, not line r.

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p.

Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p.

▼ 2.15 — C

Choice C is correct. It's given that line p is defined by the equation $2y + 8x = 11$. The equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line. Subtracting $8x$ from both sides of the equation $2y + 8x = 11$ yields $2y = -8x + 11$. Dividing both sides of this equation by 2 yields $y = -4x + \frac{11}{2}$. Therefore, the slope of line p in the xy-plane is -4. It's also given that line p is perpendicular to line r. Perpendicular lines have slopes that are negative reciprocals of each other, so the slope of line r is $-\frac{1}{-4} = \frac{1}{4}$.

Choice A is incorrect. This is the slope of line p, not line r.

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p.

Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p.

▼ 2.16 — D

Choice D is correct. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy-plane and $0, b$ is the y-intercept. Subtracting $10x$ from each side of the given equation, $10x - 5y = -12$, yields $-5y = -10x - 12$. Dividing each side of this equation by -5 yields $y = 2x + \frac{12}{5}$. This equation is in the form $y = mx + b$, where $m = 2$. Therefore, the slope of the graph of the given equation in the xy-plane is 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.17 — C

Choice C is correct. It's given that in triangle QRS, sides QR and RS each have a length of x centimeters. Therefore, the expression $2x$ represents the sum of the lengths, in centimeters, of sides QR and RS. It's also given that side SQ has a length of y centimeters. Therefore, the expression $2x + y$ represents the sum of the lengths, in centimeters, of sides QR, RS, and SQ. Since $2x + y$ is the sum of the lengths, in centimeters, of the three sides of the triangle and $2x + y = 37$, it follows that 37 is the sum of the lengths, in centimeters, of the three sides of the triangle.

Choice A is incorrect. The difference, in centimeters, between the lengths of sides QR and SQ is $x - y$, not 37.

Choice B is incorrect. The difference, in centimeters, between the lengths of sides QR and RS is $x - x$, or 0, not 37.

Choice D is incorrect. The length, in centimeters, of one of the two sides of equal length is x, not 37.

▼ 2.18 — A

Choice A is correct. The number of birds can be found by calculating the value of b when $r = 16$ in the given equation. Substituting 16 for r in the given equation yields $2.5b + 516 = 80$, or $2.5b + 80 = 80$. Subtracting 80 from both sides of this equation yields $2.5b = 0$. Dividing both sides of this equation by 2.5 yields $b = 0$. Therefore, if the business cares for 16 reptiles on a given day, it can care for 0 birds on this day.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.19 — 49

The correct answer is 49. It's given that the equation $7g + 7b = 840$ represents the number of blue tiles, b, and the number of green tiles, g, an artist needs for an 840-square-inch tile project. It's also given that the artist needs 71 blue tiles for the project. Substituting 71 for b in the equation $7g + 7b = 840$ yields $7g + 7(71) = 840$, or $7g + 497 = 840$. Subtracting 497 from both sides of this equation yields $7g = 343$. Dividing both sides of this equation by 7 yields $g = 49$. Therefore, the artist needs 49 green tiles for the project.

▼ 2.20 — 4.51, 451/100

The correct answer is 4.51. It's given that the equation $4.51x + 6.07y = 896.86$ represents this situation, where x is the number of smaller containers sold, y is the number of larger containers sold, and 896.86 is the store's total sales, in dollars, of blueberries last month. Therefore, $4.51x$ represents the store's sales, in dollars, of smaller containers, and $6.07y$ represents the store's sales, in dollars, of larger containers. Since x is the number of smaller containers sold, the price, in dollars, of each smaller container is 4.51.

▼ 2.21 — 9

The correct answer is 9. It's given that the equation $y = px + r$ defines the line. In this equation, p represents the slope of the line and r represents the y-coordinate of the y-intercept of the line. It's given that the line has a slope of 9. Therefore, the value of p is 9.

▼ 2.22 — A

Choice A is correct. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m is the slope of the line in the xy-plane that represents the relationship, and b is the y-coordinate of the y-intercept. The slope can be computed using any two points on the line. The slope of a line between any two points, x_1, y_1 and x_2, y_2, on the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. In the given table, each value of x and its corresponding value of y can be represented by a point x, y. In the given table, when the value of x is 1, the corresponding value of y is 11 and when the value of x is 2, the corresponding value of y is 16. Therefore, the points $1, 11$ and $2, 16$ are on the line. Substituting $1, 11$ and $2, 16$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{16 - 11}{2 - 1}$, or $m = 5$. Substituting 5 for m in the equation $y = mx + b$ yields $y = 5x + b$. Substituting the first value of x in the table, 1, and its corresponding value of y, 11, for x and y, respectively, in this equation yields $11 = 51 + b$, or $11 = b + 5$. Subtracting 5 from both sides of this equation yields $6 = b$. Substituting 6 for b in the equation $y = 5x + b$ yields $y = 5x + 6$. Therefore, the equation $y = 5x + 6$ represents the linear relationship between x and y.

Choice B is incorrect. For this relationship, when the value of x is 1, the corresponding value of y is 16, not 11.

Choice C is incorrect. For this relationship, when the value of x is 2, the corresponding value of y is 17, not 16.

Choice D is incorrect. For this relationship, when the value of x is 1, the corresponding value of y is 17, not 11.

▼ 2.23 — 46

The correct answer is 46. It's given that a chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL) and that the volume of acetic acid in the mixture is 10 mL. Let x represent the volume of water, in mL, in the mixture. The equation $x + 10 = 56$ represents this situation. Subtracting 10 from both sides of this equation yields $x = 46$. Therefore, the volume of water, in mL, in the mixture is 46.

▼ 2.24 — B

Choice B is correct. It's given that line *s* is parallel to line *r* in the *xy*-plane. This means that the slope of line *r* is equal to the slope of line *s*. Line *r* is defined by the equation $4x - 9y = 3$. This equation can be written in slope-intercept form $y = mx + b$, where *m* represents the slope of the line and *b* represents the *y*-coordinate of the *y*-intercept of the line. Subtracting $4x$ from both sides of the equation $4x - 9y = 3$ yields $-9y = -4x + 3$. Dividing both sides of this equation by -9 yields $y = \frac{4}{9}x - \frac{1}{3}$. Therefore, the slope of line *r* is $\frac{4}{9}$. Since parallel lines have equal slopes, the slope of line *s* is also $\frac{4}{9}$.

Choice A is incorrect. This is the reciprocal of the slope of line *s*, not the slope of line *s*.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.25 — C

Choice C is correct. An equation of a line can be written in the form $y = mx + b$, where *m* is the slope of the line and $0, b$ is the *y*-intercept of the line. The line shown passes through the point $0, -8$, so $b = -8$. The line shown also passes through the point $-8, 0$. The slope, *m*, of a line passing through two points x_1, y_1 and x_2, y_2 can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. For the points $0, -8$ and $-8, 0$, this gives $m = \frac{-8 - 0}{0 - -8}$, or $m = -1$. Substituting -8 for *b* and -1 for *m* in $y = mx + b$ yields $y = -1x + -8$, or $y = -x - 8$. Therefore, an equation of the graph shown is $y = -x - 8$.

Choice A is incorrect. This is an equation of a line with a slope of -2, not -1.

Choice B is incorrect. This is an equation of a line with a slope of 1, not -1.

Choice D is incorrect. This is an equation of a line with a slope of 2, not -1.

▼ 2.26 — .3846, 5/13

The correct answer is $\frac{5}{13}$. The graph of a line in the *xy*-plane can be represented by the equation $y = mx + b$, where *m* is the slope of the line and *b* is the *y*-coordinate of the *y*-intercept. The given equation can be written as $y = \frac{5}{13}x - 23$. Therefore, the slope of the graph of this equation in the *xy*-plane is $\frac{5}{13}$. Note that 5/13, .3846, 0.385, and 0.384 are examples of ways to enter a correct answer.

▼ 2.27 — D

Choice D is correct. An equation defining a line in the *xy*-plane can be written in the form $y = mx + b$, where m represents the slope and $0, b$ represents the *y*-intercept of the line. It's given that line t passes through the point $0, 9$; therefore, $b = 9$. The slope, m, of a line can be found using any two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $0, 9$ and $1, 17$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{17 - 9}{1 - 0}$, or $m = 8$. Substituting 8 for m and 9 for b in the equation $y = mx + b$ yields $y = 8x + 9$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.28 — C

Choice C is correct. The relationship between s and P can be modeled by a linear equation of the form P = ks + a, where k and a are constants. The table shows that P increases by 10 when s increases by 3, so k = $\frac{10}{3}$. To solve for a, substitute one of the given pairs of values for s and P: when s = 3, P = 8, so $8 = \frac{10}{3}(3) + a$, which yields a = –2. The solution is therefore $P = \frac{10}{3}s - 2$.

Choice A is incorrect. When s = 3, P = 8, but 3(3) + 10 = 19 ≠ 8. Choice B is incorrect. This may result from using the first number given for P in the table as the constant term a in the linear equation $P = ks + a$, which is true only when s = 0. Choice D is incorrect and may result from using the reciprocal of the slope of the line.

▼ 2.29 — C

Choice C is correct. It's given that the equation $4x + 6y = 36$ represents the situation where Lily filled x small containers and y large containers with all the jam she made, which was 36 cups. Therefore, $6y$ represents the total number of cups of jam in the large containers.

Choice A is incorrect. The number of large containers Lily filled is represented by y, not $6y$.

Choice B is incorrect. The number of small containers Lily filled is represented by x, not $6y$.

Choice D is incorrect. The total number of cups of jam in the small containers is represented by $4x$, not $6y$.

▼ 2.30 — 14.66, 14.67, 44/3

The correct answer is $\frac{44}{3}$. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the xy-plane and $0, b$ is the y-intercept. Distributing the $\frac{1}{3}$ in the equation $y = \frac{1}{3}29x + 10 + 5x$ yields $y = \frac{29}{3}x + \frac{10}{3} + 5x$. Combining like terms on the right-hand side of this equation yields $y = \frac{44}{3}x + \frac{10}{3}$. This equation is in the form $y = mx + b$, where $m = \frac{44}{3}$ and $b = \frac{10}{3}$. Therefore, the slope of the graph of the given equation in the xy-plane is $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

▼ 2.31 — D

Choice D is correct. An equation defining a line in the xy-plane can be written in the form $y = mx + b$, where m represents the slope and $0, b$ represents the y-intercept of the line. It's given that line k passes through the point $0, -5$; therefore, $b = -5$. The slope, m, of a line can be found using any two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $0, -5$ and $1, -1$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{-1 - -5}{1 - 0}$, or $m = 4$. Substituting 4 for m and -5 for b in the equation $y = mx + b$ yields $y = 4x - 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.32 — 1.636, 18/11

The correct answer is $\frac{18}{11}$. For a line that passes through the points x_1, y_1 and x_2, y_2 in the xy-plane, the slope of the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that a line passes through the points $4, 6$ and $15, 24$ in the xy-plane. Substituting $4, 6$ for x_1, y_1 and $15, 24$ for x_2, y_2 in the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$, yields $m = \frac{24 - 6}{15 - 4}$, or $m = \frac{18}{11}$. Therefore, the slope of the line is $\frac{18}{11}$. Note that 18/11 and 1.636 are examples of ways to enter a correct answer.

▼ 2.33 — C

Choice C is correct. The given equation $ax + by = b$ can be rewritten in slope-intercept form, $y = mx + k$, where m represents the slope of the line represented by the equation, and k represents the y-coordinate of the y-intercept of the line. Subtracting ax from both sides of the equation yields $by = -ax + b$, and dividing both sides of this equation by b yields $y = -\frac{a}{b}x + \frac{b}{b}$, or $y = -\frac{a}{b}x + 1$. With the equation now in slope-intercept form, it shows that $k = 1$, which means the y-coordinate of the y-intercept is 1. It's given that a and b are both greater than 0 (positive) and that $a < b$. Since $m = -\frac{a}{b}$, the slope of the line must be a value between -1 and 0. Choice C is the only graph of a line that has a y-value of the y-intercept that is 1 and a slope that is between -1 and 0.

Choices A, B, and D are incorrect because the slopes of the lines in these graphs aren't between -1 and 0.

▼ 2.34 — A

Choice A is correct. The *x*-intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the *x*-intercept is approximately $(158, 0)$. Therefore, if there are 0 K-type stars in the cluster, then there are approximately 158 M-type stars in the cluster. It's given that the total mass of the stars in this cluster is $127{,}882$ quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127{,}882}{158}$ quettagrams, or approximately 809.38 quettagrams. Thus, of the given choices, 811 is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.35 — 40

The correct answer is 40. It's given that $5G + 45R = 380$, where G is the number of green tokens and R is the number of red tokens won by one student and these tokens are worth a total of 380 points. Since the equation represents the situation where the student won points with green tokens and red tokens for a total of 380 points, each term on the left-hand

side of the equation represents the number of points won for one of the colors. Since the coefficient of G in the given equation is 5, a green token must be worth 5 points. Similarly, since the coefficient of R in the given equation is 45, a red token must be worth 45 points. Therefore, a red token is worth $45 - 5$ points, or 40 points, more than a green token.

▼ 2.36 — A

Choice A is correct. The value of k can be found using the slope-intercept form of a linear equation, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. The equation $ax + ky = 6$ can be rewritten in the form $y = -\frac{ax}{k} + \frac{6}{k}$. One of the given points, $(0, -3)$, is the y-intercept. Thus, the y-coordinate of the y-intercept -3 must be equal to $\frac{6}{k}$. Multiplying both sides by k gives $-3k = 6$. Dividing both sides by -3 gives $k = -2$.

Choices B, C, and D are incorrect and may result from errors made rewriting the given equation.

▼ 2.37 — C

Choice C is correct. The equation representing the linear relationship shown can be written in slope-intercept form $y = mx + b$, where m is the slope and $0, b$ is the y-intercept of the line. The line shown passes through the points $0, 6$ and $2, 0$. Given two points on a line, x_1, y_1 and x_2, y_2, the slope of the line can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $0, 6$ and $2, 0$ for x_1, y_1 and x_2, y_2, respectively, in this equation yields $m = \frac{0 - 6}{2 - 0}$, which is equivalent to $m = -\frac{6}{2}$, or $m = -3$. Since $0, 6$ is the y-intercept, it follows that $b = 6$. Substituting -3 for m and 6 for b in the equation $y = mx + b$ yields $y = -3x + 6$. Adding $3x$ to both sides of this equation yields $3x + y = 6$. Multiplying this equation by 6 yields $18x + 6y = 36$. It follows that the equation $18x + Ry = 36$, where R is a positive constant, represents this relationship.

Choice A is incorrect. The graph of this relationship passes through the point $0, 2$, not $0, 6$.

Choice B is incorrect. The graph of this relationship passes through the point $0, 2$, not $0, 6$.

Choice D is incorrect. The graph of this relationship passes through the point $-2, 0$, not $2, 0$.

▼ 2.38 — A

Choice A is correct. An equation for the graph shown can be written in slope-intercept form $y = mx + b$, where m is the slope of the graph and its y-intercept is $0, b$. Since the y-intercept of the graph shown is $0, 2$, the value of b is 2. Since the graph also passes through the point $4, 1$, the slope can be calculated as $\frac{1-2}{4-0}$, or $-\frac{1}{4}$. Therefore, the value of m is $-\frac{1}{4}$. Substituting $-\frac{1}{4}$ for m and 2 for b in the equation $y = mx + b$ yields $y = -\frac{1}{4}x + 2$. It's given that an equation for the graph shown is $y = fx + 14$. Substituting $fx + 14$ for y in the equation $y = -\frac{1}{4}x + 2$ yields $fx + 14 = -\frac{1}{4}x + 2$. Subtracting 14 from both sides of this equation yields $fx = -\frac{1}{4}x - 12$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.39 — 1.2, 6/5

The correct answer is $\frac{6}{5}$. It's given that line j is perpendicular to line k in the xy-plane. This means that the slope of line j is the opposite reciprocal of the slope of line k. For a line that passes through the points x_1, y_1 and x_2, y_2 in the xy-plane, the slope of the line can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. It's shown that line k passes through the points $(-6, 0)$ and $(0, -5)$ in the xy-plane. Substituting -6 for x_1, 0 for y_1, 0 for x_2, and -5 for y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{-5 - 0}{0 - (-6)}$, or $-\frac{5}{6}$. The opposite reciprocal of $-\frac{5}{6}$ is $\frac{6}{5}$. Therefore, the slope of line j is $\frac{6}{5}$. Note that 6/5 and 1.2 are examples of ways to enter a correct answer.

▼ 2.40 — A

Choice A is correct. The slope, m, of a line in the xy-plane can be found using two points on the line, x_1, y_1 and x_2, y_2, and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Based on the given table, the line representing the relationship between x and y in the xy-plane passes through the points $-6, n + 184$, $-3, n + 92$, and $0, n$, where n is a constant. Substituting two of these points, $-3, n + 92$ and $0, n$, for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{n - n + 92}{0 - -3}$, which is equivalent to $m = \frac{n - n - 92}{0 + 3}$, or $m = -\frac{92}{3}$. Therefore, the slope of the line that represents this relationship in the xy-plane is $-\frac{92}{3}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.41 — D

Choice D is correct. The equation of line h can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $0, b$ is the y-intercept of the line. It's given that line h contains the points $18, 130$, $23, 160$, and $26, 178$. Therefore, its slope m can be found as $\frac{160 - 130}{23 - 18}$, or 6. Substituting 6 for m in the equation $y = mx + b$ yields $y = 6x + b$. Substituting 130 for y and 18 for x in this equation yields $130 = 6(18) + b$, or $130 = 108 + b$. Subtracting 108 from both sides of this equation yields $22 = b$. Substituting 22 for b in $y = 6x + b$ yields $y = 6x + 22$. Since line k is the result of translating line h down 5 units, an equation of line k is $y = 6x + 22 - 5$, or $y = 6x + 17$. Substituting 0 for y in this equation yields $0 = 6x + 17$. Solving this equation for x yields $x = -\frac{17}{6}$. Therefore, the x-intercept of line k is $-\frac{17}{6}, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.42 — C

Choice C is correct. It's given from the graph that the points $0, 7$ and $8, 0$ lie on the line. For two points on a line, x_1, y_1 and x_2, y_2, the slope of the line can be calculated using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $0, 7$ for x_1, y_1 and $8, 0$ for x_2, y_2 in this formula, the slope of the line can be calculated as $m = \frac{0 - 7}{8 - 0}$, or $m = -\frac{7}{8}$. It's also given that the point $d, 4$ lies on the line. Substituting $d, 4$ for x_1, y_1, $8, 0$ for x_2, y_2, and $-\frac{7}{8}$ for m in the slope formula yields $-\frac{7}{8} = \frac{0 - 4}{8 - d}$, or $-\frac{7}{8} = \frac{-4}{8 - d}$. Multiplying both sides of this equation by $8 - d$ yields $-\frac{7}{8}8 - d = -4$. Expanding the left-hand side of this equation yields $-7 + \frac{7}{8}d = -4$. Adding 7 to both sides of this equation yields $\frac{7}{8}d = 3$. Multiplying both sides of this equation by $\frac{8}{7}$ yields $d = \frac{24}{7}$. Thus, the value of d is $\frac{24}{7}$.

Choice A is incorrect. This is the value of y when $x = 4$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.43 — 189/5, 37.8

The correct answer is $\frac{189}{5}$. A y-intercept of a graph in the xy-plane is a point where the graph intersects the y-axis, which is a point with an x-coordinate of 0. Substituting 0 for x in the given equation yields $\frac{30}{7} = -\frac{5y}{9} + 21$, or $0 = -\frac{5y}{9} + 21$. Subtracting 21 from both sides of this

equation yields $-21 = -\frac{5y}{9}$. Multiplying both sides of this equation by -9 yields $189 = 5y$. Dividing both sides of this equation by 5 yields $\frac{189}{5} = y$. Therefore, the *y*-coordinate of the *y*-intercept of the graph of the given equation in the *xy*-plane is $\frac{189}{5}$. Note that 189/5 and 37.8 are examples of ways to enter a correct answer.

▼ 2.44 — D

Choice D is correct. The *x*-coordinate *a* of the *x*-intercept $a, 0$ can be found by substituting 0 for *y* in the given equation, which gives $7x + 20 = -31$, or $7x = -31$. Dividing both sides of this equation by 7 yields $x = -\frac{31}{7}$. Therefore, the value of *a* is $-\frac{31}{7}$. The *y*-coordinate *b* of the *y*-intercept $0, b$ can be found by substituting 0 for *x* in the given equation, which gives $70 + 2y = -31$, or $2y = -31$. Dividing both sides of this equation by 2 yields $y = -\frac{31}{2}$. Therefore, the value of *b* is $-\frac{31}{2}$. It follows that the value of $\frac{b}{a}$ is $\frac{-\frac{31}{2}}{-\frac{31}{7}}$, which is equivalent to $\frac{31}{2}\frac{7}{31}$, or $\frac{7}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.45 — A

Choice A is correct. It is given that line ℓ is perpendicular to a line whose equation is x = 2. A line whose equation is a constant value of x is vertical, so ℓ must therefore be horizontal. Horizontal lines have a slope of 0, so ℓ has a slope of 0.

Choice B is incorrect. A line with slope $-\frac{1}{2}$ is perpendicular to a line with slope 2. However, the line with equation x = 2 is vertical and has undefined slope (not slope of 2). Choice C is incorrect. A line with slope –2 is perpendicular to a line with slope $\frac{1}{2}$. However, the line with equation x = 2 has undefined slope (not slope of $\frac{1}{2}$). Choice D is incorrect; this is the slope of the line x = 2 itself, not the slope of a line perpendicular to it.

▼ 2.46 — 210

The correct answer is 210. It's given that the equation $0.10x + 0.20y = 0.17(x + y)$ represents mixing a 10% solution and a 20% solution to obtain a 17% solution. It's also given that x represents the volume, in gallons, of the 10% solution. If 90.0 gallons of the 10% solution is used, it follows that $x = 90.0$. Substituting 90.0 for x in the given equation yields $0.10(90.0) + 0.20y = 0.17(90.0 + y)$, or $9 + 0.20y = 15.3 + 0.17y$. Subtracting $0.17y$ from both sides of this equation yields $9 + 0.03y = 15.3$. Subtracting 9 from both sides of this equation yields $0.03y = 6.3$. Dividing both sides of this equation by 0.03 yields $y = 210$. Therefore, the volume of the 20% solution that could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution is 210 gallons.

▼ 2.47 — D

Choice D is correct. It's given that line h is defined by $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$. This equation can be written in slope-intercept form $y = mx + b$, where m is the slope of line h and b is the y-coordinate of the y-intercept of line h. Adding 70 to both sides of $\frac{1}{5}x + \frac{1}{7}y - 70 = 0$ yields $\frac{1}{5}x + \frac{1}{7}y = 70$. Subtracting $\frac{1}{5}x$ from both sides of this equation yields $\frac{1}{7}y = -\frac{1}{5}x + 70$. Multiplying both sides of this equation by 7 yields $y = -\frac{7}{5}x + 490$. Therefore, the slope of line h is $-\frac{7}{5}$. It's given that line j is perpendicular to line h in the xy-plane. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Therefore, the slope of line j is the negative reciprocal of the slope of line h. The negative reciprocal of $-\frac{7}{5}$ is $\frac{-1}{-\frac{7}{5}}$, or $\frac{5}{7}$. Therefore, the slope of line j is $\frac{5}{7}$.

Choice A is incorrect. This is the slope of a line in the xy-plane that is parallel, not perpendicular, to line h.

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of $-\frac{7}{5}$.

Choice C is incorrect. This is the negative, not the negative reciprocal, of $-\frac{7}{5}$.

▼ 2.48 — ?

The correct answer is 30. The slope of a line can be found by using the slope formula, $\frac{y_2 - y_1}{x_2 - x_1}$. It's given that the slope of the line is 2; therefore, $\frac{y_2 - y_1}{x_2 - x_1} = 2$. According to the table, the points $(3,7)$ and $(k,11)$ lie on the line. Substituting the coordinates of these points into the equation gives $\frac{11 - 7}{k - 3} = 2$. Multiplying both sides of this equation by $k - 3$ gives $11 - 7 = 2(k - 3)$, or $11 - 7 = 2k - 6$. Solving for k gives $k = 5$. According to the table, the

points $(3,7)$ and $(12,n)$ also lie on the line. Substituting the coordinates of these points into $\frac{y_2-y_1}{x_2-x_1}=2$ gives $\frac{n-7}{12-3}=2$. Solving for n gives $n=25$. Therefore, $k+n=5+25$, or 30.

▼ 2.49 — 5

The correct answer is 5. It's given that the equation $10x + 15y = 85$ represents the situation, where x is the number of on-site training courses, y is the number of online training courses, and 85 is the total number of hours of training courses the apprentice has enrolled in. Therefore, $10x$ represents the number of hours the apprentice has enrolled in on-site training courses, and $15y$ represents the number of hours the apprentice has enrolled in online training courses. Since x is the number of on-site training courses and y is the number of online training courses the apprentice has enrolled in, 10 is the number of hours each on-site course takes and 15 is the number of hours each online course takes. Subtracting these numbers gives $15 - 10$, or 5 more hours each online training course takes than each on-site training course.

▼ 2.50 — 0.25, 1/4

The correct answer is $\frac{1}{4}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the xy-plane defined by this equation. It's given that line l is defined by $3y + 12x = 5$. Subtracting $12x$ from both sides of this equation yields $3y = -12x + 5$. Dividing both sides of this equation by 3 yields $y = -\frac{12}{3}x + \frac{5}{3}$, or $y = -4x + \frac{5}{3}$. Thus, the slope of line l in the xy-plane is -4. Since line n is perpendicular to line l in the xy-plane, the slope of line n is the negative reciprocal of the slope of line l. The negative reciprocal of -4 is $-\frac{1}{-4} = \frac{1}{4}$. Note that 1/4 and .25 are examples of ways to enter a correct answer.

▼ 2.51 — B

Choice B is correct. The linear relationship between x and y can be represented by an equation of the form $y - y_1 = mx - x_1$, where m is the slope of the graph of the equation in the xy-plane and x_1, y_1 is a point on the graph. The slope of a line can be found using two points on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Each value of x and its corresponding value of y in the table can be represented by a point x, y. Substituting the points $-s, 21$ and $s, 15$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{15 - 21}{s - -s}$, which gives $m = \frac{-6}{2s}$, or $m = -\frac{3}{s}$. Substituting $-\frac{3}{s}$ for m and the point $s, 15$ for x_1, y_1 in the equation $y - y_1 = mx - x_1$ yields $y - 15 = -\frac{3}{s}x - s$. Distributing $-\frac{3}{s}$ on the right-hand side of this equation yields $y - 15 = -\frac{3x}{s} + 3$. Adding 15 to each side of this equation yields $y = -\frac{3x}{s} + 18$. Multiplying

each side of this equation by s yields $sy = -3x + 18s$. Adding $3x$ to each side of this equation yields $3x + sy = 18s$. Therefore, the equation $3x + sy = 18s$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.52 — 33

The correct answer is 33. It's given in the table that the coordinates of two points on a line in the xy-plane are $(k, 13)$ and $(k + 7, -15)$. The y-intercept is another point on the line. The slope computed using any pair of points from the line will be the same. The slope of a line, m, between any two points, x_1, y_1 and x_2, y_2, on the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. It follows that the slope of the line with the given points from the table, $(k, 13)$ and $(k + 7, -15)$, is $m = \frac{-15 - 13}{k + 7 - k}$, which is equivalent to $m = \frac{-28}{7}$, or $m = -4$. It's given that the y-intercept of the line is $(k - 5, b)$. Substituting -4 for m and the coordinates of the points $(k - 5, b)$ and $(k, 13)$ into the slope formula yields $-4 = \frac{13 - b}{k - k - 5}$, which is equivalent to $-4 = \frac{13 - b}{k - k + 5}$, or $-4 = \frac{13 - b}{5}$. Multiplying both sides of this equation by 5 yields $-20 = 13 - b$. Subtracting 13 from both sides of this equation yields $-33 = -b$. Dividing both sides of this equation by -1 yields $b = 33$. Therefore, the value of b is 33.

▼ 2.53 — D

Choice D is correct. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m is the slope of the graph of this equation in the xy-plane and b is the y-coordinate of the y-intercept. The slope of a line between any two points x_1, y_1 and x_2, y_2 on the line can be calculated using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Based on the table, the graph contains the points $-18, -48$ and $7, 52$. Substituting $-18, -48$ and $7, 52$ for x_1, y_1 and x_2, y_2, respectively, in the slope formula yields $m = \frac{52 - -48}{7 - -18}$, which is equivalent to $m = \frac{100}{25}$, or $m = 4$. Substituting 4 for m, -18 for x, and -48 for y in the equation $y = mx + b$ yields $-48 = 4 \cdot -18 + b$, or $-48 = -72 + b$. Adding 72 to both sides of this equation yields $24 = b$. Therefore, $m = 4$ and $b = 24$. Substituting 4 for m and 24 for b in the equation $y = mx + b$ yields $y = 4x + 24$. Thus, the equation $y = 4x + 24$ represents the linear relationship between x and y. It's also given that the graph of the linear equation representing this relationship in the xy-plane passes through the point $\frac{1}{7}, a$. Substituting $\frac{1}{7}$ for x and a for y in the equation $y = 4x + 24$ yields $a = 4\frac{1}{7} + 24$, which is equivalent to $a = \frac{4}{7} + \frac{168}{7}$, or $a = \frac{172}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 2.54 — ?

The correct answer is 1.5. The total amount, in liters, of a saline solution can be expressed as the liters of each type of saline solution multiplied by the percent concentration of the saline solution. This gives $3(0.10)$, $x(0.25)$, and $(x+3)(0.15)$, where x is the amount, in liters, of 25% saline solution and 10%, 15%, and 25% are represented as 0.10, 0.15, and 0.25, respectively. Thus, the equation $3(0.10)+0.25x=0.15(x+3)$ must be true. Multiplying 3 by 0.10 and distributing 0.15 to $(x+3)$ yields $0.30+0.25x=0.15x+0.45$. Subtracting 0.15x and 0.30 from each side of the equation gives $0.10x=0.15$. Dividing each side of the equation by 0.10 yields $x=1.5$. Note that 1.5 and 3/2 are examples of ways to enter a correct answer.

▼ 2.55 — D

Choice D is correct. It's given that line j is perpendicular to line k and that line k is defined by the equation $x+y=0$. This equation can be rewritten in slope-intercept form, $y=mx+b$, where m represents the slope of the line and b represents the y-coordinate of the y-intercept of the line, by subtracting x from both sides of the equation, which yields $y=-x$. Thus, the slope of line k is -1. Since line j and line k are perpendicular, their slopes are opposite reciprocals of each other. Thus, the slope of line j is 1. It's given that the y-intercept of line j is $(0,3)$. Therefore, the equation for line j in slope-intercept form is $y=x+3$, which can be rewritten as $x-y=-3$.

Choices A, B, and C are incorrect and may result from conceptual or calculation errors.

▼ 2.56 — B

Choice B is correct. Two lines are perpendicular if their slopes are negative reciprocals, meaning that the slope of the first line is equal to -1 divided by the slope of the second line. Each equation in the given pair of equations can be written in slope-intercept form, $y = mx + b$, where m is the slope of the graph of the equation in the xy-plane and $0, b$ is the y-intercept. For the first equation, $5x + 7y = 1$, subtracting $5x$ from both sides gives

$7y = -5x + 1$, and dividing both sides of this equation by 7 gives $y = -\frac{5}{7}x + \frac{1}{7}$. Therefore, the slope of the graph of this equation is $-\frac{5}{7}$. For the second equation, $ax + by = 1$, subtracting ax from both sides gives $by = -ax + 1$, and dividing both sides of this equation by b gives $y = -\frac{a}{b}x + \frac{1}{b}$. Therefore, the slope of the graph of this equation is $-\frac{a}{b}$. Since the graph of the given pair of equations is a pair of perpendicular lines, the slope of the graph of the second equation, $-\frac{a}{b}$, must be the negative reciprocal of the slope of the graph of the first equation, $-\frac{5}{7}$. The negative reciprocal of $-\frac{5}{7}$ is $\frac{-1}{-\frac{5}{7}}$, or $\frac{7}{5}$. Therefore, $-\frac{a}{b} = \frac{7}{5}$, or $\frac{a}{b} = -\frac{7}{5}$. Similarly, rewriting the equations in choice B in slope-intercept form yields $y = -\frac{10}{7}x + \frac{1}{7}$ and $y = -\frac{a}{2b}x + \frac{1}{2b}$. It follows that the slope of the graph of the first equation in choice B is $-\frac{10}{7}$ and the slope of the graph of the second equation in choice B is $-\frac{a}{2b}$. Since $\frac{a}{b} = -\frac{7}{5}$, $-\frac{a}{2b}$ is equal to $-\frac{1}{2}\cdot-\frac{7}{5}$, or $\frac{7}{10}$. Since $\frac{7}{10}$ is the negative reciprocal of $-\frac{10}{7}$, the pair of equations in choice B represents a pair of perpendicular lines.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 2.57 — 24

The correct answer is 24. A line in the xy-plane can be defined by the equation $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept of the line. It's given that line l passes through the point $0, 0$. Therefore, the y-coordinate of the y-intercept of line l is 0. It's given that line l is parallel to the line represented by the equation $y = 8x + 2$. Since parallel lines have the same slope, it follows that the slope of line l is 8. Therefore, line l can be defined by an equation in the form $y = mx + b$, where $m = 8$ and $b = 0$. Substituting 8 for m and 0 for b in $y = mx + b$ yields the equation $y = 8x + 0$, or $y = 8x$. If line l passes through the point $3, d$, then when $x = 3$, $y = d$ for the equation $y = 8x$. Substituting 3 for x and d for y in the equation $y = 8x$ yields $d = 83$, or $d = 24$.

▼ 2.58 — B

Choice B is correct. Avery earns $10 per hour working at job A. Therefore, if she works a hours at job A, she will earn $10a$ dollars. Avery earns $20 per hour working at job B. Therefore, if she works b hours at job B, she will earn $20b$ dollars. The graph shown represents all possible combinations of the number of hours Avery could have worked at the two jobs to earn s dollars. Therefore, if she worked a hours at job A, worked b hours at job B, and earned s dollars from both jobs, the following equation represents the graph:

$10a+20b=s$, where s is a constant. Identifying any point (a,b) from the graph and substituting the values of the coordinates for a and b, respectively, in this equation yield the value of s. For example, the point $(16,0)$, where $a=16$ and $b=0$, lies on the graph. Substituting 16 for a and 0 for b in the equation $10a+20b=s$ yields $10(16)+20(0)=s$, or $160=s$. Similarly, the point $(0,8)$, where $a=0$ and $b=8$, lies on the graph. Substituting 0 for a and 8 for b in the equation $10a+20b=s$ yields $10(0)+20(8)=s$, or $160=s$.

Choices A, C, and D are incorrect. If the value of s is 128, 200, or 320, then no points (a,b) on the graph will satisfy this equation. For example, if the value of s is 128 (choice A), then the equation $10a+20b=s$ becomes $10a+20b=128$. The point $(16,0)$, where $a=16$ and $b=0$, lies on the graph. However, substituting 16 for a and 0 for b in $10a+20b=s$ yields $10(16)+20(0)=128$, or $160=128$, which is false. Therefore, $(16,0)$ doesn't satisfy the equation, and so the value of s can't be 128. Similarly, if $s=200$ (choice C) or $s=320$ (choice D), then substituting 16 for a and 0 for b yields $160=200$ and $160=320$, respectively, which are both false.

▼ 2.59 — ?

The correct answer is 1.5. The point $(16,0)$ corresponds to the situation where 16 cornflowers and 0 wallflowers are purchased. Since the total spent on the two types of flowers is $24.00, it follows that the price of 16 cornflowers is $24.00, and the price of one cornflower is $1.50. Note that 1.5 and 3/2 are examples of ways to enter a correct answer.

▼ 2.60 — ?

The correct answer is 1.25. The y-coordinate of the x-intercept is 0, so 0 can be substituted for y, giving $\frac{4}{5}x+\frac{1}{3}(0)=1$. This simplifies to $\frac{4}{5}x=1$. Multiplying both sides of $\frac{4}{5}x=1$ by 5 gives $4x=5$. Dividing both sides of $4x=5$ by 4 gives $x=\frac{5}{4}$, which is equivalent to 1.25. Note that 1.25 and 5/4 are examples of ways to enter a correct answer.

▼ 2.61 — 59/9, 6.555, 6.556

The correct answer is $\frac{59}{9}$. When the graph of an equation in the form $Ax + By = C$, where A, B, and C are constants, is translated down k units in the xy-plane, the resulting graph can be represented by the equation $Ax + By + k = C$. It's given that the graph of $9x - 10y = 19$ is

translated down 4 units in the xy-plane. Therefore, the resulting graph can be represented by the equation $9x - 10y + 4 = 19$, or $9x - 10y - 40 = 19$. Adding 40 to both sides of this equation yields $9x - 10y = 59$. The x-coordinate of the x-intercept of the graph of an equation in the xy-plane is the value of x in the equation when $y = 0$. Substituting 0 for y in the equation $9x - 10y = 59$ yields $9x - 100 = 59$, or $9x = 59$. Dividing both sides of this equation by 9 yields $x = \frac{59}{9}$. Therefore, the x-coordinate of the x-intercept of the resulting graph is $\frac{59}{9}$. Note that 59/9, 6.555, and 6.556 are examples of ways to enter a correct answer.

▼ 2.62 — C

Choice C is correct. It's given that the equation $3x + 5y = 32$ represents the situation where Keenan filled x small jars and y large jars with all the vegetable broth he made, which was 32 cups. Therefore, $3x$ represents the total number of cups of vegetable broth in the small jars and $5y$ represents the total number of cups of vegetable broth in the large jars.

Choice A is incorrect. The number of large jars Keenan filled is represented by y, not $5y$.

Choice B is incorrect. The number of small jars Keenan filled is represented by x, not $5y$.

Choice D is incorrect. The total number of cups of vegetable broth in the small jars is represented by $3x$, not $5y$.

▼ 2.63 — D

Choice D is correct. Each of the tables gives the same three values of x: 1, 2, and 4. Substituting 1 for x in the given equation yields $\frac{3}{5}1 + \frac{3}{4}y = 7$, or $\frac{3}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{3}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{32}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{128}{15}$. Therefore, when $x = 1$, the corresponding value of y for the given equation is $\frac{128}{15}$. Substituting 2 for x in the given equation yields $\frac{3}{5}2 + \frac{3}{4}y = 7$, or $\frac{6}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{6}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{29}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{116}{15}$. Therefore, when $x = 2$, the corresponding value of y for the given equation is $\frac{116}{15}$. Substituting 4 for x in the given equation yields $\frac{3}{5}4 + \frac{3}{4}y = 7$, or $\frac{12}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{12}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{23}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{92}{15}$. Therefore, when $x = 4$, the corresponding value of y for the given equation is $\frac{92}{15}$. The table in choice D gives x-values of 1, 2, and 4 and corresponding y-values of $\frac{128}{15}$, $\frac{116}{15}$, and $\frac{92}{15}$, respectively. Therefore, the table in choice D gives three values of x and their corresponding values of y for the given equation.

Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4} + y = 7$.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + y = 10$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4}y = 8$.

▼ 2.64 — .5, 1/2

The correct answer is $\frac{1}{2}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the xy-plane defined by this equation. It's given that line p is defined by $4y + 8x = 6$. Subtracting $8x$ from both sides of this equation yields $4y = -8x + 6$. Dividing both sides of this equation by 4 yields $y = -\frac{8}{4}x + \frac{6}{4}$, or $y = -2x + \frac{3}{2}$. Thus, the slope of line p is -2. If line r is perpendicular to line p, then the slope of line r is the negative reciprocal of the slope of line p. The negative reciprocal of -2 is $-\frac{1}{-2} = \frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

▼ 2.65 — ?

The correct answer is 74. The y-intercept of a line in the xy-plane is the ordered pair (x,y) of the point of intersection of the line with the y-axis. Since line k intersects the y-axis at the point $(0,-6)$, it follows that $(0,-6)$ is the y-intercept of this line. An equation of any line in the xy-plane can be written in the form $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept. Therefore, the equation of line k can be written as $y = mx + (-6)$, or $y = mx - 6$. The value of m can be found by substituting the x- and y-coordinates from a point on the line, such as $(2,2)$, for x and y, respectively. This results in $2 = 2m - 6$. Solving this equation for m gives $m = 4$. Therefore, an equation of line k is $y = 4x - 6$. The value of w can be found by substituting the x-coordinate, 20, for x in the equation of line k and solving this equation for y. This gives $y = 4(20) - 6$, or $y = 74$. Since w is the y-coordinate of this point, $w = 74$.